

PLC-STD-001

Photonic Logic Computing Standard

光影逻辑计算标准

Minimum Unit Definitions for Phase-Encoded Computation

Abstract

This document defines the minimum computational units, encoding standards, bus protocol, fault-tolerance requirements, and security architecture for Photonic Logic Computing (PLC) systems. It establishes the foundational terminology and interoperability specifications for phase-encoded optical computation, enabling cross-vendor implementation of PLC-compliant hardware.

本标准定义光影逻辑计算（PLC）系统的最小计算单元、编码标准、总线协议、容错规范及安全架构，建立相位编码光学计算的基础术语与互操作规范，使不同厂商的PLC兼容硬件可互操作。

Status of This Document

This document is a Standards Track draft. It has been submitted as prior art disclosure under MGOVL v2.0 and is available for review and adoption by standards bodies including IEEE Photonics Society and relevant IETF working groups. Priority is established by the SHA256 hash chain documented in the referenced repository.

1. Introduction and Motivation

Current computing paradigms require physical etching of logic into semiconductor substrates. This creates insurmountable barriers in manufacturing cost, reconfiguration speed, and computation traceability. Photonic Logic Computing addresses all three by defining logic through optical field topology rather than physical structure.

The key insight motivating this standard: an interference fringe IS a logic gate. The propagation angle of a light beam carries the logical state; vector summation of momenta implements Boolean operations; and the entire computation leaves no trace in any digital medium upon completion.

1.1 Scope

This standard applies to:

- Phase-encoded optical logic systems using liquid-crystal or equivalent reconfigurable substrates
- Optical Phase Array (OPA) based data injection systems
- Multi-channel photonic bus implementations
- Hardware security modules utilizing Physical Unclonable Functions (PUF) in optical substrates
- Compiler toolchains targeting PLC instruction sets

1.2 Terminology

Term	Abbreviation	Definition
Phase Slot	Slot	Minimum angular unit within a channel; width = Channel_span / N_slots
Guard Band	GB	Angular dead zone between adjacent channels; prevents crosstalk
Photonic Channel	PCh	Contiguous angular zone assigned to one logical signal stream
Relay Node	RN	Gain-medium node inserted when path energy falls below threshold τ
VOTE Gate	VG	Majority-vote fault-tolerant gate using cosine-similarity arbitration
Phase Healer	PH	Iterative OPA compensation loop for continuous phase drift correction
PUF Fingerprint	θ_PUF	Unique phase-offset vector derived from LC manufacturing variance
Blind Compute	BC	Computation masked by PUF key; result unrecoverable without physical device
Sovereignty Shard	SS	1/N fraction of a computation distributed to one core in an N-core array
SIS	SIS	Sovereign Instruction Set — the photonic machine-code specification

2. Minimum Computational Units

2.1 The Phase Slot — Minimum Logical Unit

A Phase Slot is the fundamental indivisible unit of information in a PLC system. It is defined by a center angle θ_center and a slot width w_slot :

$$\begin{aligned}\theta_center(channel, k) &= zone_start(channel) + (k + 0.5) \times w_slot \\ w_slot &= channel_span / N_slots\end{aligned}$$

Reference implementation:

$$\begin{aligned}N_slots &= 8 \quad (3\text{-bit resolution per channel}) \\ channel_span &= 70 \text{ deg} \quad (\text{after Guard Band allocation}) \\ w_slot &= 70 / 8 = 8.75 \text{ deg}\end{aligned}$$

Note: Implementations MAY increase N_slots to 16 (4-bit) or 32 (5-bit) provided Guard Band and crosstalk specifications are maintained.

2.2 The Photonic Channel — Signal Stream Container

A Photonic Channel is a contiguous angular zone carrying one independent signal stream. PLC-STD-001 defines four mandatory channels in the reference implementation:

Channel ID	Angular Zone	Center	Guard Band (each side)	Signal Domain
PCh-A	+5 ° to +75 °	+40 °	5 ° (boundaries: 0 ° , 80 °)	Opcode / Control
PCh-B	+95 ° to +165 °	+130 °	10 ° (boundary: 90 °)	Source Register 1
PCh-C	-165 ° to -95 °	-130 °	10 ° (boundary: 180 °)	Source Register 2
PCh-D	-75 ° to -5 °	-40 °	5 ° (boundaries: 0 ° , -80 °)	Destination / Immediate

2.3 Guard Band — Crosstalk Isolation

Guard Bands are mandatory dead zones between adjacent channels. Any optical signal detected within a Guard Band zone MUST be treated as an error and SHALL trigger a VOTE arbitration or Phase Healer correction.

Guard Band requirement:

Total Guard Band allocation = 4 × 10 deg = 40 deg

Minimum isolation: boundary angles classified as GUARD, not as channel data

Crosstalk specification: adjacent channel isolation > 20 dB

Error handling:

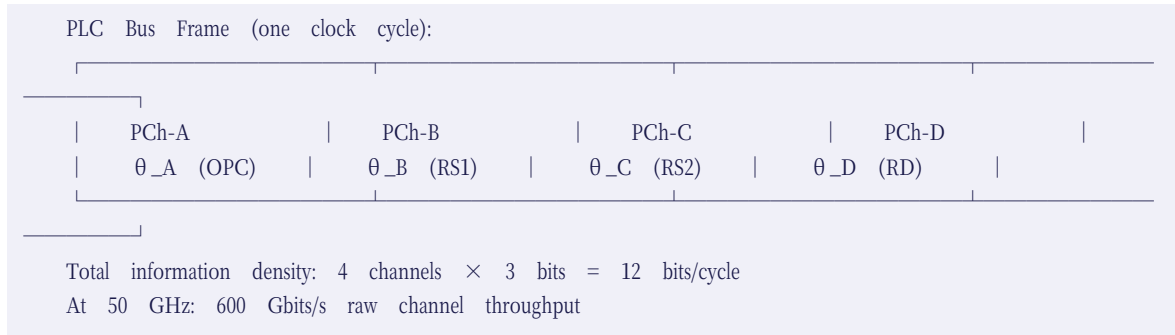
Signal in Guard Band → raise GUARD_ZONE_VIOLATION

Recovery: VOTE gate arbitration with two reference signals

3. Bus Protocol Specification

3.1 PLC Bus Frame

One PLC bus frame carries exactly one instruction per clock cycle, encoded across four simultaneous channels:



3.2 TSV-P Interconnect Parameters

Parameter	Value	Unit	Notes
Optical path pitch	50	μm	Center-to-center spacing
Core diameter	10	μm	Single-mode propagation condition
Cladding thickness	20	μm	Minimum isolation layer
Minimum bend radius	500	μm	Propagation loss < 0.1 dB/bend
Propagation delay	6.7	ps/mm	In LC medium at $\lambda = 1550\text{nm}$
Max crosstalk (adj. channel)	-20	dB	Measured at channel boundary
Clock skew budget	5	ps	Across entire bus width
P-E conversion latency	10	ps	End-to-end optical-to-electrical
Certified max frequency	200	GHz	4-phase sampling mode

4. Logic Gate Definitions

All PLC logic gates are defined by their angle-transformation rules. Implementations MUST satisfy these equations within angular tolerance $\varepsilon = 2.0^\circ$:

Gate	Inputs	Output Rule	Trigger Condition	Min Energy Output
AND	θ_A, θ_B	$\text{atan2}(\sum \sin, \sum \cos)$ of weighted vectors	Both inputs present	$\min(v_sum , (E_A + E_B)/2)$
OR	$\theta_A [, \theta_B]$	$\text{atan2}(\sum \sin, \sum \cos)$ of weighted vectors	Any input present	$(E_A + E_B)/2$
NOT	θ_A	$\theta_A + 180^\circ \pmod{360^\circ}$	Single input	$E_A \times \alpha \ (\alpha = 0.95)$
XOR	θ_A, θ_B	$\text{AND}(\text{OR}(A,B), \text{NOT}(\text{AND}(A,B)))$	Both inputs present	See cascade depth § 5
VOTE	$\theta_A, \theta_B, \theta_C$	Majority pair vector sum	3 inputs, best cosine score	$(E_maj1 + E_maj2)/2$
RELAY	—	$E_out = 1.0$ (full restore)	$E_in < \tau = 0.4$	1.0 (gain medium)

5. Energy and Cascade Depth Constraints

5.1 Measured Gate Energy Transfer Coefficients

Gate	α_{gate}	α_{eff} ($\times 0.95$ ATTN)	$N_{\text{max_safe}}$ ($\sigma=0.8$)	
AND	0.8372	0.7954	3	layers
OR	1.0000	0.9500	14	layers
NOT	0.9500	0.9025	7	layers
XOR	0.1766	0.1677	0	layers
XOR $\leftarrow$ RELAY after every				
Formula: $N_{\text{max_safe}} = \text{floor}(\sigma \times \log(\tau / E_{\text{init}}) / \log(\alpha_{\text{eff}}))$ $\sigma = 0.8$ (safety margin), $\tau = 0.4$, $E_{\text{init}} = 1.0$				

Note: Implementations MUST insert a RELAY node whenever cascade energy would fall below $\tau = 0.4$. XOR gates require RELAY insertion after every occurrence.

6. Fault Tolerance Requirements

Layer	Mechanism	Trigger	Response Time	Residual Error Spec
Layer 1	VOTE Gate	Single stray beam detected	Picosecond (gate delay)	< 5 ° angular deviation
Layer 2	Phase Healer	Drift > $\epsilon = 2.0^\circ$	Nanosecond (OPA loop)	< 2.0 ° after convergence
Layer 3	RELAY	Energy < $\tau = 0.4$	Microsecond (gain medium)	E restored to 1.0

7. Security Architecture

7.1 PUF-Based Device Identity (Mandatory for Sovereign Class)

Sovereign-class PLC implementations MUST incorporate a Liquid-Crystal PUF providing device-unique phase fingerprints. Minimum requirements:

PUF Cell Count:	≥ 64 cells
Bit Error Rate:	$< 1 \times 10^{-9}$ (standard operating conditions)
Uniqueness:	Hamming distance $\geq 45\%$ between any two devices
Mask Width:	≥ 32 bits for SIS-Secure blind computation
CRP Protocol:	MUST support Challenge-Response-Pair authentication

7.2 Sovereignty Sharding (Mandatory for Distributed Class)

Minimum cores:	$N \geq 4$
Security threshold:	Any $(N - 1)$ cores compromised $\rightarrow$ 0% information leakage
Inter-core protocol:	All inter-core signals MUST remain phase-encoded (no plaintext)
Reconstruction:	Requires all N cores' PUF masks simultaneously

8. SIS Instruction Set Reference

OPC	Mnemonic	Operation	Type	Encoding (PCh-A slot)
0	NOP	No operation	Control	Slot 0: 18.125 °
1	LOAD	$RD \leftarrow IMM$	Memory	Slot 1: 26.875 °
2	MOV	$RD \leftarrow RS1$	Register	Slot 2: 35.625 °
3	ADD	$RD \leftarrow RS1 \oplus RS2$	Arithmetic	Slot 3: 44.375 °
4	AND	$RD \leftarrow RS1 \& RS2$	Logic	Slot 4: 53.125 °
5	OR	$RD \leftarrow RS1 RS2$	Logic	Slot 5: 61.875 °
6	NOT	$RD \leftarrow \neg RS1$	Logic	Slot 6: 70.625 °
7	VOTE	$RD \leftarrow VOTE(RS1, RS2, RS3)$ Fault-Tol.		Slot 7: 79.375 °

The SIS instruction set provides Turing-complete computation when combined with memory (LOAD/STORE) and conditional branch (CMP/JGT/JEQ/JMP) extensions.

9. Compliance Levels

Level	Name	Required Features	Intended Use
PLC-L0	Observer	Read-only monitoring, no active computation	Sensors, logging
PLC-L1	Basic	AND/OR/NOT/XOR + RELAY + SIS subset	Simple edge compute
PLC-L2	Standard	All gates + full SIS + Mux/DeMux bus	General compute
PLC-L3	Resilient	L2 + 3-layer fault tolerance + Phase Healer	Critical systems
PLC-L4	Sovereign	L3 + PUF identity + SIS-Secure blind compute	Defense / Finance
PLC-L5	Distributed	L4 + $N \geq 4$ core sovereignty sharding	Classified / HFT

10. Normative References

- [PLC-PRIOR-ART] Maomao (毛广辉). Photonic Logic Chip Prior Art Record v1 – v10. github.com/maomaoati-coder/photonic-logic-chip-prior-art. 2026.
- [SHA256-CHAIN] SHA256 Hash Chain v1 – v10. See Appendix A.
- [MGOVL-V2] Maomao Governed Open Verification License v2.0. MGOVL-2026.
- [MEEP-FDTD] Oskooi et al. Meep: A flexible free-software package for electromagnetic simulations. Computer Physics Communications, 2010.
- [IEEE-802] IEEE Standard for Local and Metropolitan Area Networks. Referenced for framing conventions only.